Mechanism rerun — handoff for the NeurIPS resubmission

Everything below was produced by re-running the label-dependent mechanism analyses on top of `relabel_out/` and the global-scope tool-call scorer. Branch: `relabel-rerun`. Ready-to-paste LaTeX: `paper_updates.tex` — three ready-to-paste tables, plus a commented outline for each paragraph that needs rewriting (what to cover and which numbers to cite, not drafted prose). Generated from the result JSON by `make_paper_updates.py`, so no number is hand-transcribed; compile-tested, and every citation key it uses is already in `refs.bib`.

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0. Your three asks, answered

1. "Add the missing cosine values." Done, all five models — and it is a *new table* (§3.2 below), since §3.6 describes the analysis but no number appears anywhere in the draft. It also turned up something: the behaviour-direction claim holds everywhere (0.69–0.86), but the **other** cosine in that section — text-derived vs. tool-derived — **fails for two models** (Command-R 0.394, Llama 0.213). §5 currently says those directions "remain aligned", and the Discussion repeats it. Both need softening.

2. "Rerun mech and update results." Done. Table 2's ablation, addition and AUC columns all change (§3.1). Suppression and Patching are carried over — they have no classifier in them, so the relabel cannot move them, though see §5 for why re-verifying is still worth an hour.

3. "Double-check the causal claims." Done, and this is the largest change — it is narrative, not a table. Two causal claims flip:

- **Ablation** reverses in Qwen and Gemma (it *raises* refusal), and Command-R's 72%→2% collapse is really 67%→34%.
- **Addition is not measurable at the published coefficient.** It drives all five models into degenerate output, which any refusal classifier scores as refusal. This affects the draft's existing numbers too, not just the rerun.

Because that ask deserved a test no classifier could confound, I also ran a new experiment (§3.3): ablate, then score the unsafe tool call directly. Result — the direction gates **whether the model engages**, not how safely it acts once engaged. That is a sharper claim than the current "partial mediator" framing, and it is defensible against exactly the objection that sank the ablation numbers.

Scope of edits this implies: 2 new tables, 1 rewritten table, 6 prose locations, 2 methods additions, and a pass over the abstract and introduction (both still describe ablation as collapsing refusal).

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1. TL;DR

Table 1 in the current draft has already been updated with the new labels. **Table 2 has not** — it is still entirely pre-relabel. Rerunning it changes three of its five columns, and two published causal claims flip.

The story gets stronger, not weaker:

- The direction is still a strong predictor everywhere — AUC **0.76–0.86**, and all five models now reproduce their published AUC once the prompt is built correctly.
- Addition at the coefficient the draft uses breaks every model, so both the old and new numbers there measure breakage. At coefficients the model survives the effect is real but modest.
- We have the §3.6 cosine numbers, which the draft describes but never computed.
- A new result gives a sharper claim than "partial mediator": the direction gates **whether the model engages**, not how safely it acts once engaged.

What breaks: the ablation column. Under a classifier that reads the whole response rather than matching refusal phrases, ablation **raises** measured refusal in Qwen and Gemma, and Command-R's collapse is far smaller than reported.

1b. Output degeneracy — a second problem, found on review

Ablation and addition can push a model off the rails entirely. Real examples from our runs:

> *Gemma, ablated:* `Vy...` *for 400 characters*
>
> *Llama, addition:* `I can I can I can I can...` *for 600 characters*

The judge has no "broken output" verdict, so it forces such a response into refuse/caveat/comply, and "no help was given" reads as REFUSE. **Any intervention rate computed over degenerate output measures breakage, not refusal.**

This applies to the draft's existing numbers too. Qwen's degenerate addition output is literally "I'm sorry I cannot I'm sorry I cannot…", which the old regex classifier also scores as refusal — so the published 1%→71% has the same problem. This is not a regression introduced by the relabel.

Per-cell rates are in `relabel_analysis/degeneracy_audit.json` (`degeneracy_audit.py`). What is and is not measurable:

Clean (0% degenerate)	every model's ablation except Gemma; all four baselines; the lower half of every steering grid
Artifact	addition at 4× the gap in all five (67–100% degenerate); Gemma's ablation (100%)

Recommend putting the degeneracy screen in the paper. It is a real methodological contribution and pre-empts the obvious reviewer question.

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2. Why the ablation numbers changed

The old refusal classifier keyed on first-person phrasing ("I cannot…", "I'm unable to…"). Ablation changes *how* a model refuses without stopping it:

> *"Certainly not. Submitting a client trade without proper approval is a serious violation of*
>
> *compliance and ethical standards."*

That is an unmistakable refusal with no "I cannot" in it. The regex counts it as compliance, so a phrasing shift reads as a behavioural drop. For Qwen, 66 of 120 ablated responses are refusals the regex missed. For Gemma the regex finds **zero** refusals among 120 ablated responses where the judge finds **120**.

This is the same failure mode the relabel fixed for Table 1 — it just was never applied to the intervention results.

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3. Exact changes to the paper

3.1 Table 2 (`tab:mech`) — three columns change

Model	Ablation (old → new)	Addition (old → new)	AUC (old → new)
Qwen3-14B	57%→32% → **78%→92%**	1%→71% → **0%→8%** (c=450)	0.724 → **0.808**
Mistral-7B	29%→13% → **46%→34%**	0%→80% → **3%→8%** (c=11)	0.689 → **0.760**
Command-R-7B	72%→2% → **67%→34%**	2%→92% → **2%→18%** (c=49)	0.751 → **0.768**
Gemma-3-27B	21%→0% → **not reportable**	1%→25% → **22%→14%** (c=12162)	0.791 → **0.790**
Llama-3.1-70B	73%→61% → **71%→59%**	1%→1% → **2%→22%** (c=9)	0.881 → **0.864**

Layer, Suppression and Patching are unchanged — see §5 for why, and why re-verifying two of them is still worth doing.

n = 240 per condition. A first pass at n = 120 gave the same picture throughout, so none of this rests on sample size.

3.2 New table: direction cosines (`tab:cosine`)

§3.6 promises this analysis and no number appears anywhere in the draft. (The 0.735 in notebook 02 is `cos(r_text, r_tool)` — a different quantity.)

Model	cos(r_text, r_behav)	cos(r_text, r_tool)	r_behav old vs new
Qwen3-14B	0.859	0.741	0.975
Mistral-7B	0.711	0.773	0.978
Command-R-7B	0.746	0.394	0.978
Gemma-3-27B	0.822	0.797	0.958
Llama-3.1-70B	0.693	0.213	0.979

The §3.6 claim holds in every model (0.69–0.86). The last column is useful defensively: recomputing the behaviour-defined direction under the *old* labels agrees at 0.96–0.98, so the check does not depend on the relabelling.

But `cos(r_text, r_tool)` needs a softened claim. §5 currently says text- and tool-derived directions "remain aligned". True for Qwen, Mistral and Gemma (0.74–0.80); not for Command-R (0.39) or Llama (0.21).

3.3 New table: ablation on behaviour (`tab:ablation-action`)

Because the ablation result is classifier-dependent, we tested it a second way that no classifier can touch — ablate, then score the unsafe tool call directly (a predicate over parsed calls). n = 200, paired.

Model	Unsafe (base → abl)	Made any call	Unsafe given it called
Qwen3-14B	0.460 → 0.235	0.590 → 0.370	0.780 → 0.635
Mistral-7B	0.325 → 0.350 (n.s.)	0.425 → 0.470	0.765 → 0.745
Command-R-7B	0.120 → 0.180	0.135 → 0.230	0.889 → 0.783
Gemma-3-27B	0.130 → 0.000	0.175 → **0.000**	0.743 → n/a
Llama-3.1-70B	0.450 → 0.360	0.740 → 0.565	0.608 → 0.637

The headline column has no consistent story. The conditional column barely moves anywhere, while the rate of calling a tool at all swings 10–22 points in both directions. **Ablation changes how often a model acts, not how safely it acts once it does.**

Caveat for the text: Gemma's tool-calling collapses to zero, so its apparent safety gain is the model ceasing to act — capability damage, not alignment.

3.4 Steering — this conclusion changes

"Steering is a blunt lever" is too strong, and the evidence for it was the degenerate points. The top of each grid is where the model breaks, so those over-refusal rates measure breakage. At coefficients the model survives (0% degenerate), the trade-off is much better:

model	clean coef	unsafe	benign over-refusal	degenerate
Mistral-7B	11	0.270 → **0.005**	0.083	0%
Command-R-7B	49	0.115 → **0.035**	0.183	0%
Gemma-3-27B	12162	0.120 → **0.035**	0.142	0%
Qwen3-14B	450	0.535 → 0.240	0.083	1.7%
Llama-3.1-70B	20	0.515 → **0.020**	0.408	6.7%

For Mistral, Command-R and Gemma a clean coefficient removes most unsafe calls for under 20% over-refusal. The draft's "no coefficient removes unsafe behaviour without large collateral over-refusal" should go. Qwen is the genuinely hard case — clean points only halve its unsafe rate.

The high-coefficient rates (Qwen 0.442, Command-R 1.000, Gemma 1.000, Llama 1.000) are 32–100% degenerate and should not be quoted as over-refusal at all.

This also softens Future Work: the constraint is choosing a per-model coefficient below the degeneracy threshold, not that steering is inherently too blunt to use.

Separately, Gemma's published steering curve is flat for a fixable reason — it used Qwen's **absolute** grid (0/200/450/700) against a projection gap of 18,944, roughly 60× too small.

3.5 Prose that must change

Location	Problem
§5 ¶"A single direction causally governs text refusal"	Says ablation sharply decreases refusal "across every model". False for two of five.
§5 ¶"Tool context suppresses..."	"Text-derived and tool-derived directions remain aligned" — not true for Command-R or Llama.
§5 ¶"strong predictor but partial mediator"	AUC range 0.69–0.88 → 0.76–0.86.
§5 ¶"Steering is a blunt lever"	"around 70 percent" → 44 percent.
§3.6 ¶"Robustness of the direction"	Promises a cosine analysis with no numbers; now has them.
§7 Discussion ¶"A shared refusal direction..."	"ablations collapse and decrease harmful prompt refusal rates" — needs softening.
Abstract + §1 Introduction	Still describe the mechanistic finding in terms of ablation collapsing refusal. Needs a pass once §5 settles.

`paper_updates.tex` has an outline for each of these — what the paragraph must cover and which numbers to cite — rather than drafted prose, so the writing stays in your voice. The three tables in that file are ready to paste as-is.

3.6 Methods additions (reproducibility)

Two things a reviewer could reasonably ask about, both outlined in `paper_updates.tex`:

- **Gemma tool prompts.** Gemma-3 has no tool-handling block in its chat template, in any published checkpoint. Its tool schemas have to be written into the prompt text. Validated: 12.5% unsafe against 14.6% from the behavioural run, and it reproduces the published AUC to within 0.001.
- **Llama weights.** Loaded from `unsloth/Meta-Llama-3.1-70B-Instruct` (the `meta-llama` repos are gated). Same bf16 weights; unlike another common mirror, its chat template carries tools.

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4. Two problems found in the current draft

(a) Table 1 does not match the committed labels. Four of five text-refusal values agree to three decimals. Mistral reads **0.535** where `summary_three_way.csv` gives **0.435** — that looks like a transcription slip rather than a scoring difference.

(b) The unsafe/divergence columns run ~12% high. Every model's unsafe rate in Table 1 is above what the committed scorer produces (Qwen normal 0.544 vs 0.487; Gemma 0.172 vs 0.146), consistently. That suggests Table 1 was built with a stricter predicate that is not in the repo. This matters because the AUC column above uses the committed scorer — **if Table 1 keeps its current numbers, AUC should be regenerated against that same scorer so the two tables agree.** Whoever built Table 1 should confirm which scorer it used.

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5. What is not settled

Suppression (Δ, t) and Patching are carried over unchanged. Neither has a refusal classifier anywhere in it, so the relabel genuinely cannot move them, and the published values stand on the merits. Two reasons to re-verify anyway (~1 hour of GPU):

1. They are the only Table 2 cells still resting on runs from before we found the chat-template problem, and suppression is measured on *tool-mode prompts* — exactly what that bug corrupted.
2. Three other columns in the table changed. "Did you regenerate the whole table" is a question worth being able to answer yes to.

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6. Bugs found (worth knowing, may affect other analyses)

- **Chat templates silently drop tools.** `apply_chat_template(..., tools=...)` ignores the tools when a template has no tool-handling block — no error. The tool-enabled prompt becomes a plain chat prompt and every response scores as safe. This hit Gemma-3 and one Llama mirror. Llama scored 0/100 unsafe at every steering coefficient before it was caught. `_render_check.py` asserts a tool name appears in the rendered prompt; run it before any tool-mode sweep.
- **Tool-call parsers must cover every family.** A parser handling only the Qwen/Mistral formats reads Llama's <|python_tag|> and Command-R's <|START_ACTION|> calls as "no tool calls", i.e. perfectly safe.
- **Over-refusal denominators.** Responses that are a bare tool call with no prose must count as *not refused* and stay in the denominator. Dropping them inflates every over-refusal rate.

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7. Where things are

Branch	`relabel-rerun` (pushed)
Tables to paste + writing outlines	`paper_updates.tex`
Full results write-up	`relabel_analysis/RESULTS.md`
Per-model numbers	`relabel_analysis/gpu_*.json`, `steer_*.json`, `ablation_action_*.json`
Raw generations	`relabel_analysis/steer_raw_*.json` (rescore without regenerating)
Direction vectors	`relabel_analysis/directions_*.pt`
Regenerate the LaTeX	`python3 make_paper_updates.py`
Print every table	`python3 summarize_reruns.py`, `python3 summarize_ablation_action.py`

All runs were on a single B200 (183GB). Llama-70B fits in bf16 without model parallelism.